B. Math. III – Semestral Examination

Introduction to Differential Geometry

November 30, 2009

All Questions carry 10 marks.

1.

(a) (*Wirtinger's Inequality*) Let $F : [0, \pi] \to \mathbb{R}$ be a smooth function such that $F(0) = F(\pi) = 0$. Then, prove that

$$\int_0^\pi (\frac{dF}{dt})^2 dt \geq \int_0^\pi F(t)^2 dt,$$

with equality holding if and only if $F(t) = A\sin(t)$ for all t, where A is a constant.

(b) If a, b are positive constants, prove that

$$\int_0^{2\pi} \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}\; dt \geq 2\pi\sqrt{ab},$$

with equality holding if and only if $a = b$.
(Hint: Use Isoperimetric inequality)

2. Define conformal mapping between two surfaces. Show that the stereographic projection from the unit sphere without the north pole to the plane is a conformal map.

3. Let S be a smooth surface and C be a curve on S passing through a point P. Define the normal curvature of C at P and prove that all curves lying on S having the same tangent vector at P have the same normal curvature at P(*Meusinier's Theorem*).

4. Define an umbilical point of a smooth surface in $\mathbb{R}^3$. Classify smooth surfaces in $\mathbb{R}^3$ having only umbilical points.

5. If S is a smooth compact surface, prove that there exists a point P at which the Gaussian curvature of S is strictly positive.